



PRODUCT NAME

**Premium Dog Shampoo Formula
(Sulfate free, Silicone Free, Cold-process, Price <\$5-\$6)**

Second Review

Description

This is a liquid formulation with cold-process manufacturing to preserve the integrity of natural ingredients. It's sulfate-free for gentle cleansing and paraben-free for reduced allergenic potential. The addition of natural fragrance under 0.5% minimizes irritation risk while offering a pleasant scent. Protein and botanical conditioners enhance coat strength and shine, and the absence of silicones prevents residue build-up. There are no artificial colors to ensure safety and a natural appearance. The pet-optimized pH of 6–6.5 maintains skin and coat health, and a safe preservative system extends shelf life, making this product both effective and affordable with a cost point of <\$5-\$6.

Technical Specifications

pH Level: 6–6.5

Viscosity: Medium viscosity

Shelf Life: 12 months

Batch Size: 10–50 liters

Processing Time: 1 hours

Temperature: Process at room temperature.

Storage Conditions: Store in a cool, dry place away from moisture and direct sunlight

Equipment: Stainless steel mixing vessel, pH meter, thermometer, stirring apparatus

Formulation Table

Sr.No	Ingredient	INCI Name	%	Function
1	Water	Aqua	58.0%	Solvent
2	Cocamidopropyl Betaine	Cocamidopropyl Betaine	15.0%	Mild surfactant
3	Decyl Glucoside	Decyl Glucoside	10.0%	Gentle cleanser
4	Hydrolyzed Oat Protein	Hydrolyzed Oat Protein	3.0%	Conditioning agent
5	Aloe Vera Extract	Aloe Barbadensis Leaf Juice	5.0%	Soothing and conditioning
6	Glycerin	Glycerin	4.0%	Humectant
7	Panthenol	Panthenol	2.0%	Moisturizing agent
8	Natural Fragrance	Natural Fragrance	0.5%	Natural fragrance
9	Citric Acid	Citric Acid	0.5%	pH adjuster
10	Potassium Sorbate	Potassium Sorbate	1.0%	Preservative
11	Sodium Benzoate	Sodium Benzoate	1.0%	Preservative
	Total		100%	

Manufacturing Process

Phase A

1. Combine Water (Aqua) and Glycerin in a mixing tank.
2. Stir until the mixture is homogenous.

Phase B

1. Add Cocamidopropyl Betaine and Decyl Glucoside to Phase A.
2. Mix gently until fully blended, avoiding excessive foam formation.

Phase C

1. Incorporate Hydrolyzed Oat Protein, Aloe Vera Extract, and Panthenol into mix.
2. Stir thoroughly until all ingredients are well integrated.

Phase D

- 1-Add Natural Fragrance, Citric Acid, Potassium Sorbate, and Sodium Benzoate.
- 2-Mix gently ensuring the oil is dispersed evenly.
- 3-Check the pH level and adjust with Citric Acid to reach a pH of 6–6.5.

Required Equipment

Stainless steel mixing vessel, pH meter, thermometer, stirring apparatus

Safety Precautions

Handling: Wear appropriate PPE including gloves, safety glasses, lab coat.

PPE Requirements: Chemical-resistant gloves, safety goggles, protective clothing.

Storage: Store in cool, dry place away from direct sunlight. Keep containers tightly

closed. **Storage Conditions:** Store in a cool, dry place away from moisture and direct light.

Packaging Notes

Packaging: Use chemically compatible containers (HDPE, glass, or PET).

Labeling: Include product name, ingredients, usage instructions, and safety warnings.

Certification: Complies with natural and sulfate-free formulation standards

Scaling Note

Lab Scale: This formulation is designed for laboratory testing and development.

Pilot Scale: For pilot production, scale proportionally and verify all parameters.

Production Scale: Consider equipment limitations, mixing efficiency, and process validation.

Batch Size: Current formulation is optimized for 10–50 liters.

Scaling Factor: Maintain ingredient ratios while adjusting processing parameters as needed.

Product Usage Instructions & Application Guidelines

Wet the dog's coat thoroughly with water. Apply the shampoo in a moderate amount, starting from the neck and working downwards. Massage gently into the fur to lather, avoiding the eyes and ears. Rinse thoroughly with clean water and towel dry.